

ADDON 3: Algebraic Wavefront Acceleration, Polynomial Extrapolation, and Finite Field Co-Processors

11.4 Polynomial Matrix Acceleration and Vector Extrapolation

To further bypass the execution latency of iterative transformation loops within ultra-long context windows, the ADAM-PentaD architecture natively supports hardware-level matrix polynomial mappings and vector extrapolation protocols. Rather than executing explicit, sequential iteration steps for convergence inside the analog core, the VLIW compiler maps the computational state transitions to minimized Chebyshev matrix polynomials. This directly minimizes the spectral radius of the physical propagation operator, yielding bounded convergence rates without explicit full-matrix materialization.

Furthermore, the architecture integrates a dedicated vector extrapolation pipeline executing Minimal Polynomial Extrapolation (MPE) and Reduced Rank Extrapolation (RRE) directly on the host digital plane (Core Alpha/Beta). By analyzing a shallow trajectory of successive historical token embedding vectors, the system analytically predicts the steady-state convergence point of local hidden layer activations. This reduces the number of physical wavefront settling cycles required during complex multi-step reasoning chains, effectively flattening iterative deep learning trajectories into direct $O(1)$ geometric projections.

11.5 Finite Field (Galois) Co-Processing and Group-Theoretic Dimensionality Reduction

To mitigate the inherent thermal noise floor and Analog-to-Digital (ADC) power walls of continuous complex arithmetic, the architecture introduces an auxiliary Galois Field ($GF(2^m)$) computational mode. By mapping structured, deterministic sub-matrices entirely into finite field representations, the system executes discrete, error-free matrix operations via modular arithmetic. This completely eliminates floating-point rounding Overheads and quantization errors in high-entropy cryptographic and token-routing layers. This finite-field mapping acts as a hardware-native Reed-Solomon layer, providing Physical Byzantine Fault Tolerance (PBFT) and instant data encryption directly within the routing cells.

Concurrently, the compiler exploits Group Representation Theory to optimize dense tensor computations. When input datasets or structural weights possess spatial or permutation symmetries (e.g., $SO(3)$ rotations or cyclic groups), the Clifford-Algebra compiler decomposes the monolithic matrix space into a block-diagonal configuration of low-dimensional irreducible representations.

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graph LR; A["[Dense Matrix Input]"] --> B["[Group Representation Filter]"]; B --> C["[Parallel Block-Diagonal MVM]"]; C --- D["├─ Block 1 (Clifford Rotors)"]; C --- E["├─ Block 2 (Complex Spectra)"]; C --- F["└─ Block 3 (Galois Modular)"];
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This structural factorization transforms a singular, computationally prohibitive Matrix-Vector Multiplication (MVM) into a series of highly parallelized, independent operations executed across small, power-efficient analog sub-crossbars, maximizing throughput on resilient legacy fabrication nodes.

11.6 Physical Realizability and Co-Design Constraints

While abstract algebraic formulations offer substantial theoretical complexity reductions, the physical implementation within ADAM-PentaD is bounded by the layout efficiency of standard-cell standard EDA toolchains (Yosys/nextpnr). To prevent look-up table (LUT) explosion and transistor-level routing congestion within 28-nm/65-nm/90-nm++ legacy silicon, Galois multipliers and polynomial accumulator loops are restricted to fixed bit-widths ($m \leq 8$). The polynomial and finite-field primitives are co-designed alongside the physical characteristics of the photonic-memristive lattice, ensuring that mathematical symmetry reductions translate directly into real-world energy drops ($\sim 10^{-6}$ Wh per token) rather than control-path overhead.